Executive summary

What the 1,000 GLP-1 abstracts revealed

The experiment uses a frozen set of 1,000 English PubMed abstracts. Titles and keywords are excluded; four cumulative preprocessing conditions share identical source texts. The values below are measured results, not textbook expectations.

Condition	Vocabulary	Zipf k	R^2	RMSE log10
A	28,792	1.0915	0.9571	0.1003
B	19,231	1.2508	0.9687	0.0975
C	19,115	1.2095	0.9633	0.1024
D	15,569	1.2663	0.9613	0.1101

1. Zipf is an approximation

Condition B is strongly linear on log-log axes (R^2=0.9687), but its estimated k=1.2508 differs from 1. Head, middle and tied-frequency tail behave differently. R^2 does not establish a power law; regional fits and residuals reveal the limits of the approximation.

2. Preprocessing changes the empirical answer

Punctuation normalization reduces vocabulary from 28,792 to 19,231. Stopword removal reduces tokens to 177,052; Porter reduces vocabulary further to 15,569. A single exponent without the preprocessing settings would hide these differences.

3. Frequency is not document coverage

CF counts occurrences; DF counts distinct documents. In condition B, 'and' occurs 10,988 times across 997 abstracts, with IDF=0.0030. In contrast, 'cdr' has CF=15, DF=1 and CF/DF=15.00: repetition is concentrated rather than widespread.

4. Implications and limits

The corpus supports explicit decisions about stopwords, dictionary size, postings and IDF weighting. It does not establish better retrieval accuracy or measured compression savings. The useful lesson beyond the textbook is the size and direction of preprocessing effects in this particular domain, and the need to inspect CF and DF separately. A fixed snapshot, public PMID list and exported settings make the conclusions reproducible.